

Pseudocode

After determining the number of nodes in the hidden layer (and, in other applications, whether there will be 1 or 2 hidden layers)

Stochastic Gradient Descent

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1   create a network with the required number of input, hidden and
    output nodes. This mostly amounts to creating a List of Lists of
    thetas for the hidden and output layers. Don't forget biases for
    every node.
2   initialize all thetas to small random values (0..1) or (-1, 1).
3   until termination
4       for each point in the training set
5           # feed forward step
6           calculate output of every node in the network.
7           # back prop step
8           calculate delta_o for every output node
9           calculate delta_h for every hidden node
10          update all of the thetas
```

Comments

As was discussed in the lecture, this is still basically gradient descent except we updated the thetas after each observation instead of with the average gradient. Note: if you have trouble with the *stochastic* part, go back to “batch” gradient descent as in the last assignment.

With stochastic gradient descent, you generally work through all of the data points before you do your termination check. The termination check is the same one we've been using if the change in error falls below some epsilon. You'll need to add that code to the pseudocode.

The "magic" is the back propagation of the error.

The equations for the above code are:

$$\delta^o \leftarrow \hat{y}(1 - \hat{y})(y - \hat{y})$$

$$\delta^h \leftarrow \hat{y}(1 - \hat{y}) \sum \theta^{i,o} \delta^{i,o}$$

remember that the summation above disappears if there is only one output node.

$$\theta_j \leftarrow \theta_j + \alpha \delta^i x_j$$

where the δ is from the next layer and the x is the signal from the previous layer.